MTH 201: Probability and Statistics

Mid Semester Exam (Section B)

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Sanjit K. Kaul

No books, notes, or devices are allowed. Just a pen/ pencil and eraser. Institute rules will apply with regards to cheating. Show your steps. Don't prioritize simplifying products and sums, if you are short on time. You have about 120 minutes.

Question 1. 10 **marks** There are two coins. One of the coins, on tossing, gives heads (H) with probability 0.4, tails (T) with probability 0.4, and gives an indeterminate result (I) otherwise. The other coin gives heads (H) with probability 0.5 and tails (T) otherwise. A coin is chosen randomly from the two coins. The chosen coin is tossed 5 times. The five coin tosses result in the sequence $HHHTT$. Given the observation, calculate your revised beliefs (conditional probabilities) about which coin was chosen to toss 5 times.

Question 2. 20 **marks** A coin toss outputs 1 with probability 0.6 and outputs 0 otherwise. Consider 5 coin tosses. Answer the following questions.

1) (8 marks) Suppose the 5 coin tosses result in an even sum. Derive the conditional expected value of the sum of coin tosses.
2) (8 marks) Suppose the 5 coin tosses result in an odd sum. Derive the conditional expected value of the sum of coin tosses.
3) (4 marks) Use the conditional expected values obtained in the two parts above to calculate the expected value (unconditional) of the sum of coin tosses.

Question 3. 20 **marks** (5 marks) A student begins preparing for the mid semester examination of *Probability and Statistics* a night before the exam. Given the impossibility of working through enough word problems, the student decides to try and commit formulas in the textbook to memory. Any attempt by the student to commit a formula to memory is successful with probability q. Derive the PMF of the random variable that counts the number of attempts including the one in which the student successfully commits the kth formula to memory. Assume outcomes of all attempts are independent.

(15 marks) Suppose the student successfully commits to memory the 3rd formula in the 5th attempt. Derive the conditional PMF of the number of attempts the student took to successfully commit to memory the first formula. Derive the corresponding conditional expected value.

Question 4. 50 **marks** A coin toss results in 1 with probability p and 0 otherwise. An experiment procedure has the coin tossed n times. Let the random variables $X_1, X_2, \ldots, X_n$ be the outcomes of, respectively, the 1st, 2nd,..., n^{th} coin tosses. The random variable $S_n = \sum_{i=1}^{n} X_i$ is the sum of the outcomes of the n coin tosses.

The outcomes of the n coin tosses are processed by n multiplier circuits to output the n random variables $M_1, M_2, \ldots, M_n$. Specifically, the ith multiplier circuit outputs the random variable $M_i = X_1 \times X_2 \times \ldots \times X_i$. The outputs of the multiplier circuits are summed to give $Q_n = \sum_{i=1}^{n} M_i$. Answer the following questions.

(a) (5 marks) Derive the PMF of the random variable S_n.

(b) (15 marks) Derive the PMF, the expected value, and the variance of the random variables M_i, $i = 1, \ldots, n$.

(c) (15 marks) Derive the PMF and expected value of Q_n.

(d) (15 marks) Derive the conditional PMF of Q_n, given $X_1 = 1$. Also derive the corresponding conditional expected value.